

AWS checklist

Station name	QAS M	
People present	NBK, NSF, PILOT	instructed by: NSF
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	24/8-2017 9 ¹⁶	

Notes:

WATER IN LOGGER ENCLOSURE

Part	Part number upon arrival	Part number upon departure, if replaced	Check after maintenance:
Radiometer			- Clean? - Aligned with inclinometer and mast?
Inclinometer			- Aligned with radiometer and mast?
Satellite antenna			
Wind sensor			- Black box pointing true south (i.e. along the mast)?
Temperature / humidity assembly			- Fan spins and sounds OK?
HygroClip in T/hum assembly			- NB: could have a 40 °C offset
Sonic ranger on AWS			- Old sensor with new membrane: Y/N
Solar panel			- Clean? - NB: is not effective when partially covered
GPS antenna			- Stuck to top of logger enclosure?
Data logger			
Multiplexer			- Clicking during measurement cycle?
Iridium modem IMEI number			- Transmits?
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - NB: make sure not being buried next winter
Thermistor string			
Ablation hose assembly			- No air in hose and bladder half full?
Logger enclosure	- Replaced (including everything inside): Y/N - New desiccant bags: Y/N		
Battery box	- New box or new batteries: Y/N - New desiccant bags: Y/N		
Stakes	- New ones drilled in: Y/N - Length of new stakes 8 m: Y/N - Caps in bottom end to reduce melt-in: Y/N		

Metadata BEFORE maintenance (if establishing a new AWS: skip this)

Time difference between logger clock and UTC	+ / -
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Screen dump of values in fast scan mode	Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	Y / N
Radiometer aligned with mast and inclinometer? If not, measure.	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	287 cm
Sonic ranger on stakes height (membrane – surface)	380 cm
Free length of stake – outer, holding boom with sonic ranger	416 cm
Free length of stake – middle	437 cm
Free length of stake – outer, holding boom without sonic ranger	420 cm
Depth of ablation hose using markings on hose	725 cm
Height of ablation hose fluid level above the ice surface	1005 152 cm
Vertical difference in surface height (at station – at hose) (approx.)	-40 cm
Length of thermistor string on surface from surface marking	300 cm
Stake assembly still standing?	☒ Y / N / sort of
Apparent damage to sensors? If yes, which? <i>logger</i>	☒ Y / N

Metadata AFTER maintenance / establishment

Time logger clock set to UTC?	Y / N
Name of logger program	
Screen dump of values in fast scan mode	Y / N
Latitude/longitude (if new or relocated)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north (radiometer should point to true south)	° true/magn
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Depth of ablation hose using markings on hose	— cm
Height of ablation hose fluid level above the ice surface	— cm
Vertical difference in surface height (at station – at hose) (approx.)	— cm
Length of thermistor string on surface from surface marking	-35 cm